

CS6650 Assignment 3: Database Design Document

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Database Choice Justification

Amazon DynamoDB Selection

Key Advantages:

- **Parallel Query Execution:** Distributed architecture enables concurrent queries across partitions
- **Horizontal Scaling:** Automatic partitioning and scaling without manual intervention
- **Write-Heavy Optimization:** Superior performance for high-throughput message ingestion
- **Future-Proof:** Flexible schema supports multimedia content expansion

vs. PostgreSQL: PostgreSQL requires complex sharding, manual partitioning, and has single-master write limitations. DynamoDB auto-scales seamlessly and handles distributed writes more effectively.

Trade-off: Some queries slower (unique user counting) but acceptable for superior write performance and scaling.

Schema Design

Primary Table: `chatflow-messages`

```
json
{
  "messageId": "550e8400-e29b-41d4-a716-446655440000",
  "roomId": 15,
  "userId": "user12345",
  "username": "john_doe",
  "message": "Hello everyone!",
  "timestamp": 1764025941604,
  "messageType": "TEXT"
}
```

Primary Key Design:

- **Partition Key:** `roomId` (Number) - Distributes writes evenly across rooms

- **Sort Key:** `timestamp` (Number) - Enables chronological ordering within rooms

Key Design Decisions:

- `roomId` partitioning prevents hot partitions and enables sub-100ms room queries
- `timestamp` sort key supports efficient time-range queries
- Document structure allows easy multimedia field additions
- UUID `messageId` ensures uniqueness across distributed system

Indexing Strategy

Index Name	Keys	Query Support	Performance
Primary	PK: roomId, SK: timestamp	Room messages by time range	30-57ms
user-timestamp-index	PK: userId, SK: timestamp	User message history	5-7ms
room-timestamp-index	PK: roomId, SK: timestamp	Room query flexibility	30-57ms
type-timestamp-index	PK: messageType, SK: timestamp	Message type analytics	<100ms

Query-Index Mapping

Query 1: Room Messages (Target: <100ms)

- **Index:** Primary Key (roomId, timestamp)
- **Rationale:** Direct partition access with sort key range eliminates cross-partition scanning
- **Result:** 30-57ms for 1000 messages

Query 2: User History (Target: <200ms)

- **Index:** user-timestamp-index GSI
- **Rationale:** User messages span multiple rooms; GSI provides single-partition user view
- **Result:** 5-7ms across all rooms

Query 3: Active Users Count (Target: <500ms)

- **Index:** Parallel table scan with timestamp filter (8 segments)
- **Rationale:** No single index can optimize unique user counting across partitions; parallel scan distributes workload

- **Optimization:** Multi-threaded parallel scan with 8 segments reduces execution time from 24+ seconds to under 500ms
- **Result:** 390-630ms (acceptable for infrequent analytics, 48x performance improvement)

Query 4: User's Rooms (Target: <50ms)

- **Index:** user-timestamp-index with room aggregation
 - **Rationale:** Query user partition, extract unique roomIds efficiently
 - **Result:** 5-7ms
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Scaling Considerations

Auto-Scaling Architecture

- **Write Capacity:** Auto-scales 5-40,000 units based on demand
- **Partition Management:** Automatic splitting at 10GB boundaries
- **Global Distribution:** Multi-region replication with Global Tables

Performance Characteristics

- **Write Scaling:** Linear improvement with partition count
- **Read Scaling:** GSI partitions scale independently
- **Query Performance:** Room/user queries maintain <100ms at any scale

Capacity Planning

- **Expected Growth:** Linear message volume, exponential user growth
 - **Scaling Triggers:** Write throttling events trigger auto-scaling
 - **Resource Management:** Automatic without operational overhead
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Backup and Recovery

Built-in Features

- **Point-in-Time Recovery:** 35-day continuous backups
- **Multi-AZ Replication:** 99.999999999% durability across 3 AZs
- **Cross-Region Backup:** Global Tables for disaster recovery

Recovery Strategy

- **Automated Monitoring:** CloudWatch alarms for performance degradation
 - **Failover Procedures:** Cross-region failover with minimal RTO/RPO
 - **Data Integrity:** Strong consistency for backup operations
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Future Considerations

Schema Evolution

- **Multimedia Support:** S3 integration for large binary content
- **Message Types:** Easy addition of IMAGE, VIDEO, FILE types
- **Advanced Features:** Message threading, reactions with separate GSIs

Performance Optimization

- **Read Caching:** DynamoDB Accelerator (DAX) for sub-millisecond reads
- **Write Optimization:** Batch processing and connection pooling
- **Global Scale:** Multi-region active-active deployment